

MICHAŁ BALCEREK

Assistant Professor | Mathematics

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Faculty of Pure and Applied Mathematics

Department of Applied Mathematics · Hugo Steinhaus Center

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RESEARCH PROFILE

Research on stochastic processes, anomalous diffusion and statistical inference, connecting models of memory, heterogeneity and spatial dependence with observed trajectories. Applications include single-particle tracking in biological systems, environmental time series, and financial and actuarial models.

- **Random and evolving memory:** fractional and multifractional Brownian motion, random Hurst exponents, telegraphic multifractional models, switching dynamics and ergodicity.
- **Multidimensional diffusion:** dependent spatial components, covariance and spectral analysis, turning-angle statistics and anisotropy in molecular trajectories.
- **Statistical identification:** testing fractional models, discriminating Gaussian processes with quadratic-form statistics, assessing Gaussianity and accounting for measurement noise.
- **Trajectory analysis:** change-point detection, parameter estimation, hidden Markov models and the collaborative CINNAMON approach combining statistical and neural-network methods.
- **Applied stochastic modelling:** heterogeneous membrane and cytoplasmic diffusion, environmental measurements, subordinated option-pricing models and machine learning for catastrophe bonds.

ACADEMIC APPOINTMENTS

2017–present	Wrocław University of Science and Technology Academic staff; currently Assistant Professor (<i>adiunkt</i>), Faculty of Pure and Applied Mathematics.
Feb 2021– Feb 2022	Postdoctoral Researcher · University of Birmingham, UK School of Computer Science. Statistical and machine-learning methods for biological experimental data.

EDUCATION

May 2018	PhD in Mathematics Wrocław University of Science and Technology, Faculty of Pure and Applied Mathematics. Thesis: <i>Stochastic and numerical modelling of signals having anomalous diffusion behaviour</i> . Supervisor: Prof. Krzysztof Burnecki.
2012	MSc in Mathematics Wrocław University of Science and Technology. Specialisation: Financial and Actuarial Mathematics. Thesis: <i>Risk process with different loss distributions</i> . Supervisor: Prof. Agnieszka Wyłomańska.
2010	Bachelor's degree in Mathematics · Wrocław University of Science and Technology.

RESEARCH FUNDING AND PROJECTS

- 2026/27 **Fulbright Senior Award 2026** · Polish-U.S. Fulbright Commission
Data-Driven Fractional Random Anomalous Motion. Awarded for a seven-month research stay at Colorado State University with Diego Krapf, scheduled for 2026/27.
 Fulbright Senior Award; the programme is currently named Fulbright Polish Scholar Award, from the 2027 edition.
- 2023–2024 **Principal Investigator, NCN MINIATURA 7**
Analysis of doubly stochastic processes with long memory. National Science Centre, Poland; grant 2023/07/X/ST1/01139.
- 2018 **University research grant** · Wrocław University of Science and Technology
Investigation and development of methods for modelling processes with anomalous diffusion. Targeted research funding, reference 0402/0053/18.
- 2013–2018 **Investigator, NCN MAESTRO** · PI: Aleksander Weron
Anomalous dynamics of complex physical and biological systems: stochastic modeling and statistical identification. Grant 2012/06/A/ST1/00258.
- Project period:
 2013–2016 **Investigator, NCN OPUS 4** · PI: Andrzej Szczurek
The variability of physical and chemical parameters in time as a source of comprehensive information about indoor air quality. Grant 2012/07/B/ST8/03031.

RESEARCH VISITS AND COLLABORATIONS

- May–Aug 2024 **Colorado State University, Fort Collins, USA**
 Research stay in Diego Krapf’s laboratory through NCN MINIATURA 7. Doubly stochastic processes with long memory, ergodic properties, statistical estimators and measurement noise.
- Jan 2022 **Colorado State University, Fort Collins, USA**
 Visiting researcher in Diego Krapf’s group.
- Apr 2014 **University of Würzburg, Germany**
 Research visit.
- Research collaborations have included the groups of Diego Krapf (Colorado State University), Carlo Manzo (University of Vic, Spain), and Davide Calebiro (then at the University of Würzburg). The Birmingham postdoctoral appointment and planned Fulbright stay are listed above.

HONOURS AND DISTINCTIONS

- 2026 **Secundus programme, seventh edition** · Laureate in Mathematics, for research output published in 2025.
- 2021, 2023,
 2024, 2025 **Rector’s Prizes** · Teaching and scientific activities.
- 2024 **Quintus Prize** · Student-selected distinction for teaching excellence.
- 2022–2024 **Academia Iuvenum** · Member of the interdisciplinary programme for early-career researchers.
- All distinctions above are from Wrocław University of Science and Technology. The Fulbright award is described under research funding.

SELECTED PUBLICATIONS

- [1] **M. Balcerek**, A. Pacheco-Pozo, A. Wyłomańska, K. Burnecki, and D. Krapf. Two-dimensional fractional Brownian motion: analysis in time and frequency domains. *Journal of Physics A: Mathematical and Theoretical* 59(15), 155202 (2026). doi: 10.1088/1751-8121/ae59d1.
- [2] **Preprint. M. Balcerek**, A. Pacheco-Pozo, A. Wyłomańska, and D. Krapf. Turning angle analysis reveals hidden anisotropies in the anomalous diffusion of molecules in live cells. 2026. arXiv:2608.07975.
- [3] **M. Balcerek**, A. Pacheco-Pozo, A. Wyłomańska, K. Burnecki, and D. Krapf. Two-dimensional Brownian motion with dependent components: turning angle analysis. *Chaos* 35(2), 023166 (2025). doi: 10.1063/5.0227369.
- [4] **M. Balcerek**, A. Pacheco-Pozo, A. Wyłomańska, and D. Krapf. Evaluating Gaussianity of heterogeneous fractional Brownian motion. *Journal of Physics A: Mathematical and Theoretical* 58(17), 175001 (2025). doi: 10.1088/1751-8121/adc9e5.
- [5] **M. Balcerek**, S. Thapa, K. Burnecki, H. Kantz, R. Metzler, A. Wyłomańska, and A. Chechkin. Multifractional Brownian motion with telegraphic, stochastically varying exponent. *Physical Review Letters* 134(19), 197101 (2025). doi: 10.1103/PhysRevLett.134.197101.
- [6] **Preprint. J. Kończal, M. Balcerek**, and K. Burnecki. Machine learning models for predicting catastrophe bond coupons using climate data. 2025. arXiv:2512.22660.
- [7] J. W. Malinowski, M. R. Kostrzewa, **M. Balcerek**, W. Tomczuk, and J. Szwabiński. CINNAMON: a hybrid approach to change point detection and parameter estimation in single-particle tracking data. *Journal of Physics: Photonics* 7(3), 035008 (2025). doi: 10.1088/2515-7647/add826.
- [8] **Correction. M. Balcerek**, K. Burnecki, S. Thapa, A. Wyłomańska, and A. Chechkin. Erratum: “Fractional Brownian motion with random Hurst exponent: accelerating diffusion and persistence transitions” [*Chaos* 32, 093114 (2022)]. *Chaos* 34(4), 049901 (2024). doi: 10.1063/5.0210418.
- [9] **M. Balcerek**, G. P. Krzyżanowski, and M. Magdziarz. About subordinated generalizations of 3 classical models of option pricing. *Mathematica Applicanda* 52(1), 3–25 (2024). doi: 10.14708/ma.v52i1.7256.
- [10] A. Pacheco-Pozo, **M. Balcerek**, A. Wyłomańska, K. Burnecki, I. M. Sokolov, and D. Krapf. Langevin equation in heterogeneous landscapes: how to choose the interpretation. *Physical Review Letters* 133(6), 067102 (2024). doi: 10.1103/PhysRevLett.133.067102.
- [11] **M. Balcerek**, A. Wyłomańska, K. Burnecki, R. Metzler, and D. Krapf. Modelling intermittent anomalous diffusion with switching fractional Brownian motion. *New Journal of Physics* 25(10), 103031 (2023). doi: 10.1088/1367-2630/ad00d7.
- [12] W. Wang, **M. Balcerek**, K. Burnecki, A. V. Chechkin, S. Janušonis, J. Ślęzak, T. Vojta, A. Wyłomańska, and R. Metzler. Memory-multi-fractional Brownian motion with continuous correlations. *Physical Review Research* 5(3), L032025 (2023). doi: 10.1103/PhysRevResearch.5.L032025.
- [13] **M. Balcerek**, K. Burnecki, S. Thapa, A. Wyłomańska, and A. V. Chechkin. Fractional Brownian motion with random Hurst exponent: accelerating diffusion and persistence transitions. *Chaos* 32(9), 093114 (2022). doi: 10.1063/5.0101913.

SELECTED PUBLICATIONS (CONTINUED)

- [14] **M. Balcerek**, K. Burnecki, G. Sikora, and A. Wyłomańska. Discriminating Gaussian processes via quadratic form statistics. *Chaos* 31(6), 063101 (2021). doi: 10.1063/5.0044878.
- [15] J. Janczura, **M. Balcerek**, K. Burnecki, A. Sabri, M. Weiss, and D. Krapf. Identifying heterogeneous diffusion states in the cytoplasm by a hidden Markov model. *New Journal of Physics* 23(5), 053018 (2021). doi: 10.1088/1367-2630/abf204.
- [16] **M. Balcerek** and K. Burnecki. Testing of fractional Brownian motion in a noisy environment. *Chaos, Solitons & Fractals* 140, 110097 (2020). doi: 10.1016/j.chaos.2020.110097.
- [17] **M. Balcerek** and K. Burnecki. Testing of multifractional Brownian motion. *Entropy* 22(12), 1403 (2020). doi: 10.3390/e22121403.
- [18] K. Maraj, D. Szarek, G. Sikora, **M. Balcerek**, A. Wyłomańska, and I. Jabłoński. Measurement instrumentation and selected signal processing techniques for anomalous diffusion analysis. *Measurement: Sensors* 7–9, 100017 (2020). doi: 10.1016/j.measen.2020.100017.
- [19] D. Szarek, G. Sikora, **M. Balcerek**, I. Jabłoński, and A. Wyłomańska. Fractional dynamics identification via intelligent unpacking of the sample autocovariance function by neural networks. *Entropy* 22(11), 1322 (2020). doi: 10.3390/e22111322.
- [20] **M. Balcerek**, H. Loch-Olszewska, J. A. Torreno-Pina, M. F. Garcia-Parajo, A. Weron, C. Manzo, and K. Burnecki. Inhomogeneous membrane receptor diffusion explained by a fractional heteroscedastic time series model. *Physical Chemistry Chemical Physics* 21(6), 3114–3121 (2019). doi: 10.1039/C8CP06781C.
- [21] A. Szczurek, M. Maciejewska, A. Wyłomańska, G. Sikora, **M. Balcerek**, and M. Teuerle. Normal and anomalous diffusion in fluctuations of dust concentration nearby emission source. *Physica A: Statistical Mechanics and its Applications* 491, 619–631 (2018). doi: 10.1016/j.physa.2017.08.146.
- [22] G. Sikora, E. Kepten, A. Weron, **M. Balcerek**, and K. Burnecki. An efficient algorithm for extracting the magnitude of the measurement error for fractional dynamics. *Physical Chemistry Chemical Physics* 19(39), 26566–26581 (2017). doi: 10.1039/C7CP04464J.
- [23] A. Weron, K. Burnecki, E. J. Akin, L. Solé, **M. Balcerek**, M. M. Tamkun, and D. Krapf. Ergodicity breaking on the neuronal surface emerges from random switching between diffusive states. *Scientific Reports* 7, 5404 (2017). doi: 10.1038/s41598-017-05911-y.
- [24] A. Szczurek, M. Maciejewska, A. Wyłomańska, G. Sikora, **M. Balcerek**, and M. Teuerle. Discrimination of particulate matter emission sources using stochastic methods. *Physica A: Statistical Mechanics and its Applications* 463, 452–466 (2016). doi: 10.1016/j.physa.2016.06.071.
- [25] **Book chapter. M. Balcerek** and A. Weron. Stochastic dynamics of G-protein-coupled cell-surface receptors. In: A. Steland, E. Rafajłowicz, and K. Szajowski (eds.) *Stochastic Models, Statistics and Their Applications*. Springer Proceedings in Mathematics & Statistics **122**. Springer International Publishing (2015), pp. 233–240. doi: 10.1007/978-3-319-13881-7_26.

TALKS, CONFERENCE PRESENTATIONS AND ACTIVITIES

- Aug 2026* **International Conference on Trends and Perspectives in Linear Statistical Inference (LinStat), Poznań, Poland**
When switching fractional Brownian motion becomes non-Gaussian.
Invited talk.
- Jul 2026* **Mathematics in Finance, Skukuza, South Africa**
Modeling Markets with Evolving Memory.
Talk.
- Jun–Jul 2026* **23rd ECMI Conference on Industrial and Applied Mathematics, Kaunas, Lithuania**
Beyond Independent Diffusion: Modeling Correlated Multidimensional Anomalous Dynamics.
Talk; chair of a session in the minisymposium “Fractional dynamics and applications”.
- Dec 2025* **CFE–CMStatistics, London, United Kingdom**
Multidimensional fractional Brownian motion: From construction to analysis.
Talk.
- Jul 2025* **44th Conference on Stochastic Processes and their Applications, Wrocław, Poland**
Two-dimensional fractional Brownian motion: construction and analysis.
Talk.
- Jun 2025* **ANDi+: AI for Bioimaging Beyond Trajectory Analysis, Gothenburg, Sweden**
From Theory to Application: Modeling Complex Systems with Telegraphic Multifractional Brownian Motion.
Talk.
- Dec 2024* **Recent Advances in Applied Mathematics (RAAM), Wrocław, Poland**
Evaluating Gaussianity of heterogeneous fractional Brownian motion.
Talk; organiser.
- Sep 2024* **Fluctuations in Small Complex Systems VII, Venice, Italy**
2d Brownian motion with dependent components.
Poster.
- Feb 2024* **European Physical Society Statistical and Nonlinear Physics Prizes Conference, Wrocław, Poland**
Intermittent anomalous diffusion.
Talk; organiser.
- Dec 2023* **CMStatistics, Berlin, Germany**
Modelling intermittent anomalous diffusion with switching fractional Brownian motion.
Talk; chair of “Stochastic processes and applications”.
- Sep 2023* **51st Conference on Applications of Mathematics, Kościelisko, Poland**
Telegraphic multifractional Brownian motion and its applications.
Talk.
- Jun 2023* **22nd ECMI Conference on Industrial and Applied Mathematics, Wrocław, Poland**
Doubly stochastic fractional Brownian motion.
Talk; organiser.
- Dec 2022* **XLVIII Conference on Mathematical Statistics, Będlewo, Poland**
Fractional Brownian motion with random Hurst exponent.
Talk; organiser.
- Nov 2022* **Mathematics for Society: Health, Industry and Sustainable Development, Gdańsk, Poland**
Applications of anomalous diffusion and long-memory processes.
Invited talk.

TALKS, CONFERENCE PRESENTATIONS AND ACTIVITIES (CONTINUED)

- Apr 2021* **21st ECMI Conference on Industrial and Applied Mathematics, Wuppertal, Germany (online)**
Testing of Multifractional Brownian Motion.
Talk.
- Dec 2019* **Stochastic and AI Modeling of Complex Systems, Wrocław, Poland**
Testing of fractional Brownian motion in a noisy environment.
Talk.
- Nov 2019* **University of Verona, Italy**
Anomalous diffusion and applications to molecular biology.
Invited lecture series.
- Sep 2018* **Joint Meeting of the Italian Mathematical Union, the Italian Society of Industrial and Applied Mathematics and the Polish Mathematical Society, Wrocław, Poland**
Ergodicity surfaces for Gaussian processes.
Talk; organiser.
- Jul 2018* **ECMI Modelling Week, Novi Sad, Serbia**
Diffusion and anomalous diffusion models: simulation and application to biological data.
Instructor and student-team leader.
- Jun 2018* **Humboldt-Universität zu Berlin, Germany**
Statistics of Financial Markets II.
Invited lecture series; invited by Wolfgang Härdle and Alla Petukhina.
- Sep 2017* **Summer School on Normal and Anomalous Stochastic Processes in Physics, Acre, Israel**
Extracting the magnitude of the measurement error for fractional dynamics.
Poster.
- Jun 2017* **Ulam Computer Simulations Workshop, Lviv, Ukraine**
Calibration surface approach to the real data.
Poster.
- Sep 2016* **29th Marian Smoluchowski Symposium on Statistical Physics: Nonequilibrium and Theory of Fluctuations, Zakopane, Poland**
Analysing ergodicity in biological data.
Poster.
- Jul 2016* **Summer School on Mathematical Methods of Statistics, Angers, France**
Identification methods of changes in variance in processes.
Talk.
- Feb 2015* **Workshop on Stochastic Models, Statistics and Their Applications, Wrocław, Poland**
Stochastic Dynamics of G-Protein-Coupled Cell-Surface Receptors.
Talk.
- Sep 2014* **27th Marian Smoluchowski Symposium on Statistical Physics: Randomness in Nonequilibrium Systems, Zakopane, Poland**
Stochastic analysis of indoor air quality data.
Poster.
- Jun–Jul 2014* **11th International Vilnius Conference on Probability Theory and Mathematical Statistics, Vilnius, Lithuania**
Stochastic dynamics of single particle tracking biological data.
Poster.

TEACHING AND RESEARCH SUPERVISION

- Since 2013* **University teaching** · Wrocław University of Science and Technology
 Doctoral teaching duties in 2013–2017; subsequent teaching as a faculty member.
 Courses in stochastic-process simulation, numerical methods, financial engineering, stochastic modelling of financial markets, risk modelling, non-life insurance mathematics, mathematical and statistical software, probability, differential equations, calculus, real and complex analysis, and linear algebra.
- Since 2018* **Completed degree projects**
 Supervised more than 50 BSc/BEng and MSc thesis projects in applied mathematics, stochastic processes, statistical inference, computational methods and data analysis. Completed projects include the 2025/26 academic year.
 Project record: michalbalcerek.github.io/students/.
- Doctoral mentoring**
 Assistant-supervision experience with Michał Wronka (Industrial Doctorate programme) and Julia Kończal.
- Actuarial education**
 Experience organising actuarial-examination preparation meetings.

OTHER ACADEMIC ACTIVITIES

Journal peer review

Reviewed manuscripts for *Chaos, Solitons & Fractals*, *Chaos, Computational Statistics*, *Open Mathematics*, *Mathematica Applicanda*, *Fractal and Fractional*, and *Entropy*.

Conference organising committees

- Applied Mathematics Now and Beyond (2025).
- Recent Advances in Applied Mathematics (2024).
- European Physical Society Statistical and Nonlinear Physics Prizes Conference (2024).
- 22nd ECMI Conference on Industrial and Applied Mathematics (2023).
- XLVIII Conference on Mathematical Statistics (2022).
- Joint Meeting of the Italian Mathematical Union, the Italian Society of Industrial and Applied Mathematics, and the Polish Mathematical Society (2018).

University and student community

- Representative of academic teachers on the Faculty of Pure and Applied Mathematics Council.
- Former advisor of KNIF – Koło Naukowe Inżynierii Finansowej (Financial Engineering Student Interest Group).

ACTUARIAL QUALIFICATIONS AND EXPERIENCE

- 2014* **State actuarial examinations**
 Passed all four actuarial examinations organised by the Polish Financial Supervision Authority (KNF); final examination passed in 2014.
- Jul–Nov 2012* **TU Europa Group, Wrocław** · Actuarial and risk-management experience
 Assistant to the Chief Risk Officer and actuarial-department internship, including work on Solvency II implementation.

COMPUTATIONAL TOOLS AND LANGUAGES

Research and teaching tools: Julia, Python, MATLAB, R, Mathematica, SAS and $\text{\LaTeX}$.

Languages: Polish (native), English (advanced), German (beginner).